



CREDIT CARD DEFAULT PREDICTION FOR IDFC FIRST BANK

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Submission Report

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1 Executive Summary

Credit-card default prediction is a fundamentally imbalanced classification problem: while the majority of customers maintain their repayment obligations, a relatively small proportion eventually default. From a banking perspective, however, these minority cases are disproportionately important because incorrectly identifying a high-risk customer can result in direct credit losses, while incorrectly flagging a low-risk customer can result in unnecessary credit restrictions and customer friction.

This project develops a machine-learning pipeline to predict the probability that an existing credit-card customer will default, represented by the binary target variable `bad_flag`.

The development dataset contains 96,806 customer accounts and 1,214 raw predictive attributes, in addition to the account identifier and target variable. The feature space is divided into four logical families: onus attributes, transaction attributes, bureau attributes, and bureau-enquiry attributes. These feature groups differ substantially in their structure and missing-value patterns, making a single preprocessing strategy unsuitable.

The project addresses four primary modelling challenges:

1. **Severe class imbalance:** only 1.42% of customers are labelled as defaulters, creating an approximately 70:1 imbalance between non-default and default observations.
2. **High dimensionality:** the dataset contains more than 1,200 raw attributes.
3. **Heterogeneous feature families:** transaction, bureau and onus variables exhibit different statistical characteristics.
4. **Model evaluation under imbalance:** accuracy alone can be highly misleading when the positive class is rare.

The modelling pipeline therefore uses family-wise preprocessing and dimensionality reduction. Missing values are imputed, features are standardized, and five principal components are extracted independently from each of the four feature families, reducing the original 1,212 modelling variables to 20 principal components.

To address class imbalance, SMOTE (Synthetic Minority Over-sampling Technique) is applied exclusively to the training data. Multiple models are subsequently benchmarked, including Logistic Regression, Random Forest, Gradient Boosting and Decision Tree models.

The final model is a soft-voting ensemble combining Logistic Regression, Random Forest and Gradient Boosting. On the untouched test set, the ensemble achieves:

- ROC-AUC: 0.8088
- PR-AUC: 0.0736
- Recall: 71.17%

- Precision: 3.84%
- F1-score: 0.0729

The results demonstrate an important characteristic of credit-risk modelling: the model can identify a substantial proportion of actual defaulters, but doing so at the selected 0.50 threshold produces a high number of false positives. This makes threshold selection and downstream risk-policy design as important as model selection itself.

The project therefore treats the machine-learning model not as a binary “approve/reject” engine, but as a risk-ranking and early-warning mechanism that could support differentiated interventions such as enhanced monitoring, credit-line review, or targeted customer outreach.

The analysis also identifies clear avenues for improvement, particularly replacing PCA with more interpretable feature selection, using class-weighted tree boosting, performing probability calibration, optimizing thresholds against explicit business costs, and evaluating the model using precision-recall and top-risk-decile metrics.

2 Problem Statement

2.1 Business Context

A credit-card issuer must continuously manage the risk associated with its existing customer portfolio. The central question is not simply:

“Will this customer default?”

Instead, a more useful operational question is:

“Which customers are most likely to default, and how should limited risk-management resources be prioritized across them?”

A predictive credit-risk system can enable the institution to identify high-risk customers before default occurs. Such a system can support proactive actions including:

- enhanced account monitoring,
- credit-limit review,
- targeted repayment reminders,
- customer outreach,
- risk-based portfolio segmentation,
- early intervention for deteriorating accounts.

The objective of this project is therefore to develop a predictive model that estimates the probability of default from historical customer-level information.

2.2 Target Variable

The target variable is `bad_flag`. It is binary:

- 0 $\rightarrow$ customer did not default
- 1 $\rightarrow$ customer defaulted

The development dataset contains:

- 95,434 non-default accounts
- 1,372 default accounts

Therefore:

- Default rate = 1.42%
- Non-default rate = 98.58%

This produces an approximate 70:1 class imbalance.

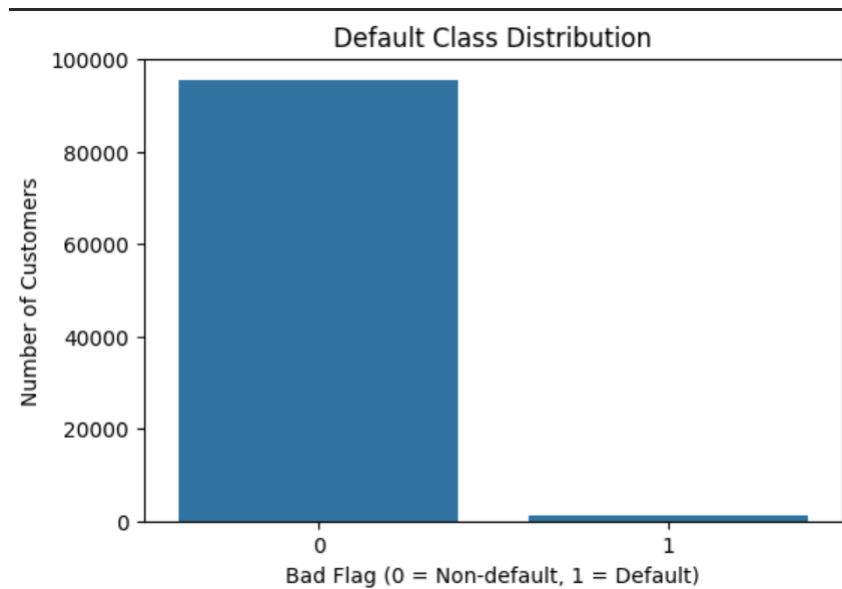


Figure 1: Default Class Distribution

The severe class imbalance means that a model predicting every customer as non-default would achieve approximately 98.6% accuracy while providing no useful risk-detection capability. Consequently, accuracy was treated as a secondary metric, while ROC-AUC, PR-AUC, precision and recall were used to assess discriminatory performance.

3 Dataset Overview and Exploratory Data Analysis

3.1 Dataset Structure

The development dataset contains:

$$96,806 \text{ accounts} \times 1,216 \text{ columns}$$

These consist of:

- 1 account identifier,
- 1 target variable,
- 1,214 raw feature variables.

After removing two completely empty columns, the dataset contains 1,214 columns.

After separating the target and removing the account identifier from the modelling matrix, the final feature matrix contains:

$$96,806 \text{ observations} \times 1,212 \text{ features}$$

3.2 Feature Families

The 1,212 modelling features are divided into four logical families.

Table 1: Feature Family Overview

Feature family	No. of features	Primary information
Onus attributes	48	Customer/account-level internal attributes
Transaction attributes	664	Transactional activity and spending behaviour
Bureau attributes	450	External credit-history information
Bureau enquiry attributes	50	Credit enquiry behaviour
Total	1,212	

This segmentation is important because the four families exhibit different missing-value patterns and statistical behaviour.

Onus Attributes

The 48 onus variables capture internal account/customer information. These variables contain missing observations and therefore require imputation before downstream modelling.

Transaction Attributes

The transaction block is the largest feature family, with 664 variables. These variables represent transaction-related behaviour and contain substantial missingness and zero-valued observations. The large number of transaction variables creates both computational and statistical challenges.

Bureau Attributes

The 450 bureau variables contain information derived from external credit history. These variables are useful from a risk perspective because prior credit behaviour can provide signals about future repayment risk.

Bureau Enquiry Attributes

The 50 bureau-enquiry variables capture credit enquiry behaviour. This family ultimately exhibits the highest concentration of variance after PCA among the four groups.

4 Data Preparation and Leakage-Safe Preprocessing

A central design principle of the pipeline was to ensure that information from the test set did not influence preprocessing decisions.

The data was split using an 80:20 stratified train-test split.

Training set

- 77,444 observations
- 1,098 defaults
- Default rate $\approx 1.42\%$

Test set

- 19,362 observations
- 274 defaults
- Default rate $\approx 1.42\%$

Stratification ensured that the severe class imbalance was preserved across both partitions. This is particularly important in credit-risk modelling because an unstratified split could produce materially different default rates between training and test data.

4.1 Family-Wise Preprocessing

Instead of treating the 1,212 variables as a single homogeneous matrix, preprocessing was performed independently for each feature family.

For each family:

1. Missing values were imputed.
2. Features were standardized.
3. PCA was fitted using training data.
4. The transformed representation was applied to both training and test data.

This design reduces the risk of leakage because the preprocessing parameters are learned from the training data and subsequently applied to unseen observations.

5 Dimensionality Reduction Using PCA

The original dataset contains 1,212 modelling features. Training conventional models directly on this feature space can create several issues:

- computational overhead,
- correlated variables,
- noisy predictors,
- increased model complexity,
- greater risk of overfitting.

Principal Component Analysis was therefore used separately within each feature family. For each family, the first five principal components were retained. This produced:

Table 2: PCA Variance Explained by Feature Family

Feature family	Original features	Components retained	Variance explained
Onus	48	5	53.45%
Transaction	664	5	13.32%
Bureau	450	5	31.54%
Bureau enquiry	50	5	69.31%
Total	1,212	20	—

The resulting modelling matrix therefore contains only 20 PCA-derived features.

5.1 Interpretation of PCA Results

The PCA results reveal an important structural difference between the feature families. The bureau-enquiry family achieves the highest variance retention, with the first five components explaining 69.31% of its variance. The onus family follows with 53.45%. The transaction family is substantially more diffuse: its first five components explain only 13.32% of total variance.

This suggests that transaction behaviour contains information distributed across a much larger number of dimensions, rather than being concentrated in a small number of latent components.

The result creates an important modelling trade-off. PCA dramatically reduces dimensionality, but the five-component representation of transaction data necessarily discards a significant amount of variance. This becomes one of the most important limitations of the current modelling approach and an obvious area for future experimentation.

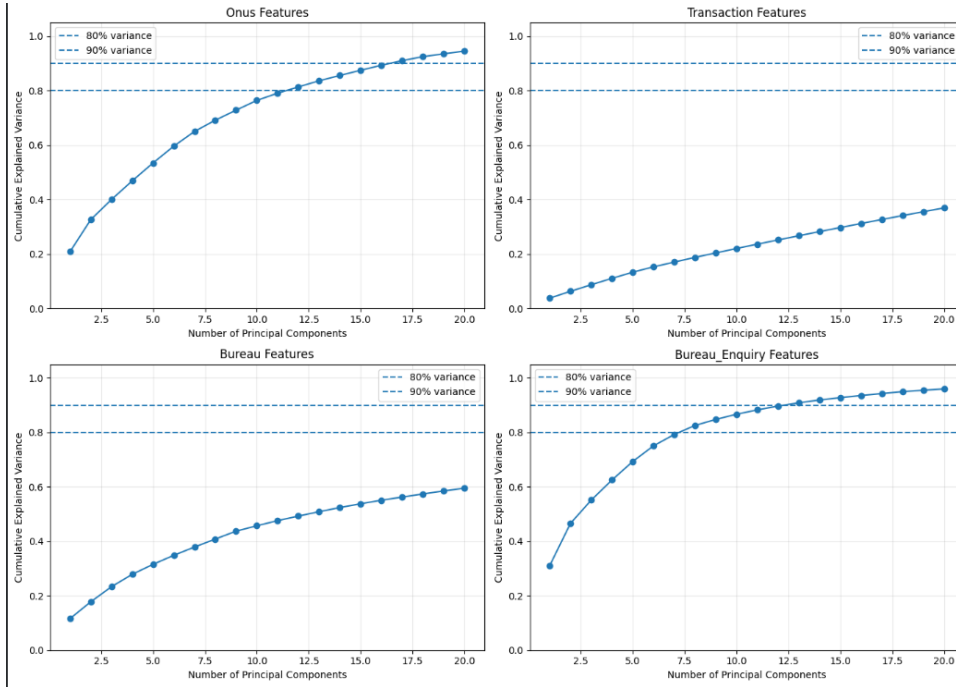


Figure 2: Cumulative explained variance across principal components for the four feature families. The first five components explain 53.45% of onus variance, 13.32% of transaction variance, 31.54% of bureau variance and 69.31% of bureau-enquiry variance.

The different variance-retention profiles indicate that the four feature families possess materially different levels of dimensional concentration. In particular, transaction attributes retain only 13.32% of their variance in the first five components, suggesting that future iterations could benefit from family-specific component counts rather than applying the same dimensionality reduction budget to every feature group.

That sentence is particularly useful in an interview because it demonstrates that you are not blindly accepting PCA as “feature engineering.”

6 Handling Severe Class Imbalance

The default class represents only approximately 1.42% of observations. Training directly on the original distribution creates a strong incentive for a model to predict the majority class. The first Logistic Regression benchmark demonstrates this problem clearly.

Without SMOTE:

- Accuracy = 98.57%
- Precision = 0%
- Recall = 0%
- F1 = 0
- ROC-AUC = 0.8080
- PR-AUC = 0.0641

The 98.57% accuracy initially appears strong. However, the model identifies none of the defaulting customers at the selected classification threshold. This demonstrates why accuracy is inappropriate as the primary metric for this problem.

6.1 SMOTE

To improve minority-class learning, SMOTE was applied only to the training set.

Before SMOTE:

- Non-default = 76,346
- Default = 1,098

After SMOTE:

- Non-default = 76,346
- Synthetic/default class = 76,346

The test set was left untouched. This is essential because the test set must represent the real-world operating distribution.

The objective of SMOTE was therefore not to make the overall dataset artificially balanced, but to provide the training algorithms with sufficient minority-class examples to learn the characteristics of default.

7 Model Development

Multiple classification algorithms were benchmarked. The objective was not simply to identify the model with the highest accuracy, but to understand the trade-off between:

- discriminatory power,
- minority-class recall,
- precision,
- model complexity,
- effect of SMOTE.

The following models were evaluated:

1. Logistic Regression
2. Logistic Regression + SMOTE
3. Random Forest + SMOTE
4. Gradient Boosting + SMOTE
5. Decision Tree + SMOTE

8 Model Benchmarking

The results on the untouched test set are shown below.

	Model	Accuracy	Precision	Recall	F1	ROC-AUC	PR-AUC
0	Logistic Regression - No SMOTE	0.985694	0.000000	0.000000	0.000000	0.807968	0.064077
1	Logistic Regression + SMOTE	0.734686	0.039052	0.751825	0.074248	0.807942	0.061047
3	Gradient Boosting + SMOTE	0.690476	0.034511	0.773723	0.066074	0.793138	0.055848
2	Random Forest + SMOTE	0.788503	0.040414	0.613139	0.075829	0.790440	0.058946
4	Decision Tree + SMOTE	0.702303	0.032049	0.686131	0.061238	0.759200	0.036542

Figure 3: Model Benchmarking Results (Test Set)

8.1 What the Results Show

The benchmark produces an important result. SMOTE substantially increases the ability of the models to identify defaulters. For example, Logistic Regression moves from:

0% recall $\rightarrow$ 75.18% recall

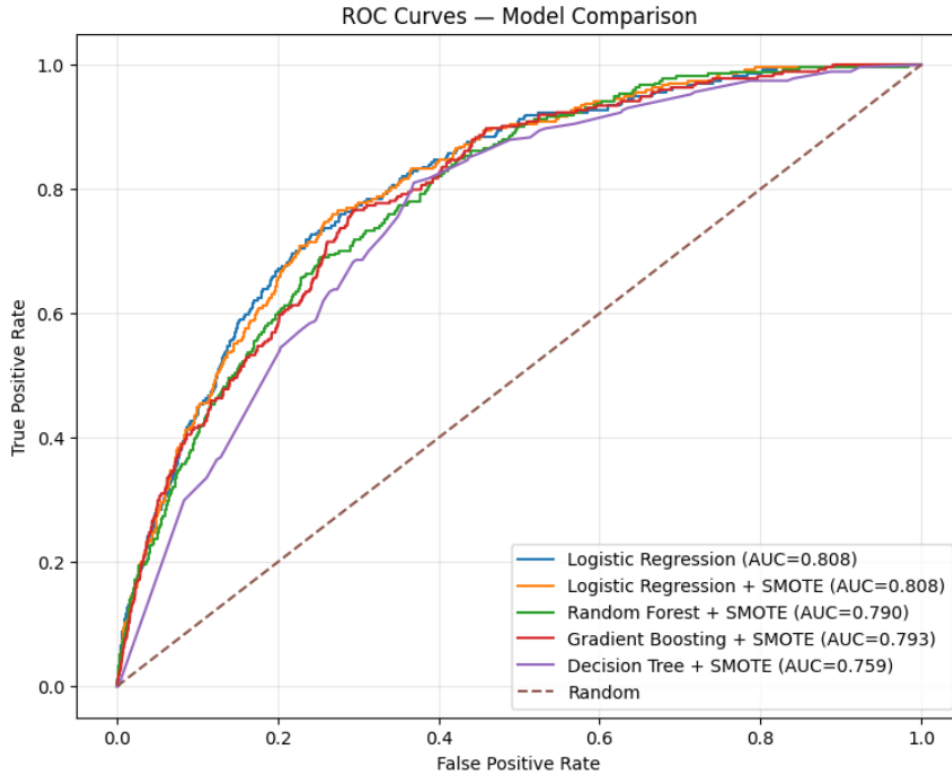


Figure 4: Model Benchmarking ROC-AUC

after applying SMOTE.

However, this improvement comes at a substantial precision cost. The SMOTE-based Logistic Regression model achieves approximately 3.9% precision, meaning that a large majority of customers flagged at the default threshold are false positives.

This is not necessarily a failure of the model. Rather, it reflects the fundamental economics of highly imbalanced credit-risk prediction: increasing recall requires the system to accept more false positives.

8.2 Why PR-AUC Matters

ROC-AUC remains useful as a broad measure of ranking/discrimination. However, for this problem, PR-AUC is especially informative because the positive class is extremely rare. A model can achieve a relatively strong ROC-AUC while still producing weak precision when identifying the minority class.

Therefore, model selection should consider:

- PR-AUC,
- recall,
- precision,
- top-risk population performance,

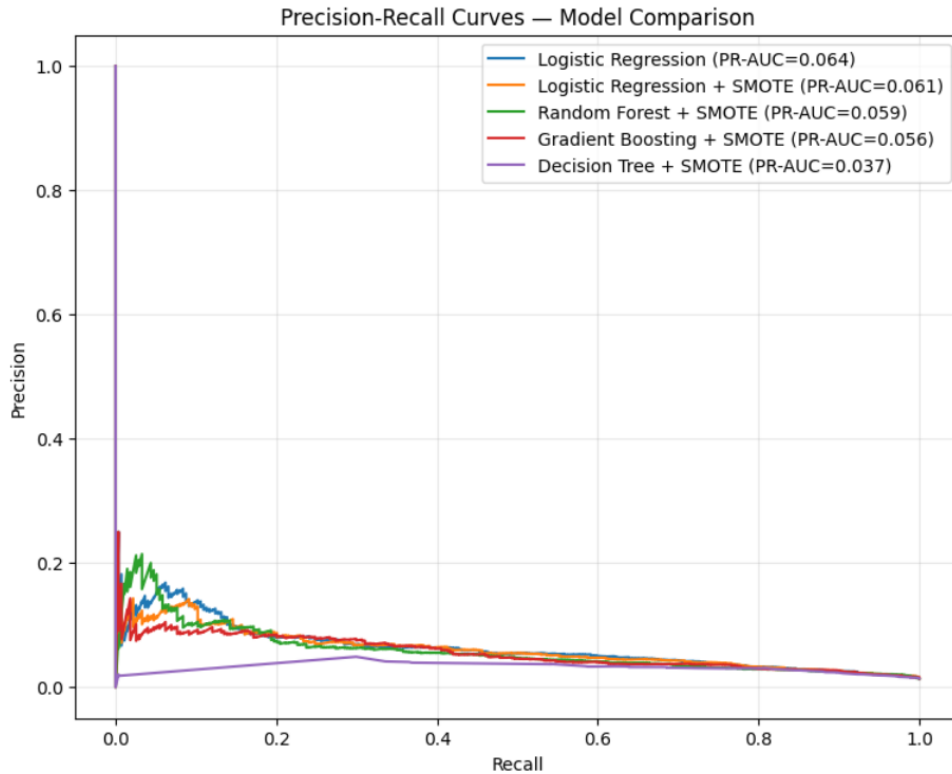


Figure 5: Model Benchmarking PR-AUC

- threshold-specific business costs.

The relatively similar ROC curves indicate that model architecture alone does not create a dramatic improvement in discriminatory performance under the current 20-dimensional PCA representation. This motivated the use of an ensemble to combine complementary model behaviour rather than relying on a single classifier.

9 Precision-Recall Analysis

The Precision-Recall curves provide a more informative view of model behaviour under the severe class imbalance.

The graph demonstrates that precision declines rapidly as recall increases. This is an important operational characteristic. A risk team that wants to identify a larger proportion of future defaulters must accept a larger number of false positives.

This creates a direct policy trade-off:

Higher recall $\rightarrow$ broader intervention population $\rightarrow$ lower precision

versus

Higher precision $\rightarrow$ narrower intervention population $\rightarrow$ missed high-risk customers

This trade-off is arguably more relevant to a bank than raw accuracy.

10 Soft-Voting Ensemble

The final model combines three complementary classifiers:

- Logistic Regression + SMOTE
- Random Forest + SMOTE
- Gradient Boosting + SMOTE

A soft-voting ensemble combines the probability predictions of these models rather than relying on a single hard classification.

The resulting ensemble achieves:

Table 3: Final Soft-Voting Ensemble Performance

Metric	Ensemble
Accuracy	74.37%
Precision	3.84%
Recall	71.17%
F1	0.0729
ROC-AUC	0.8088
PR-AUC	0.0736

The ensemble achieves the strongest PR-AUC among the models evaluated in the notebook. Its ROC-AUC of 0.8088 also marginally exceeds the best individual model's ROC-AUC.

10.1 Why Ensemble?

The motivation for ensembling was to reduce dependence on the inductive bias of any single model. The three models approach the classification problem differently:

Logistic Regression

Provides a relatively simple linear decision boundary and serves as an interpretable baseline.

Random Forest

Captures non-linear interactions through an ensemble of decision trees.

Gradient Boosting

Sequentially builds weak learners to improve errors made by earlier learners.

Combining their probability estimates creates a more robust aggregate prediction.

Figure 6: ROC curve of the final soft-voting ensemble on the held-out test set. The ensemble achieves ROC-AUC = 0.8088.

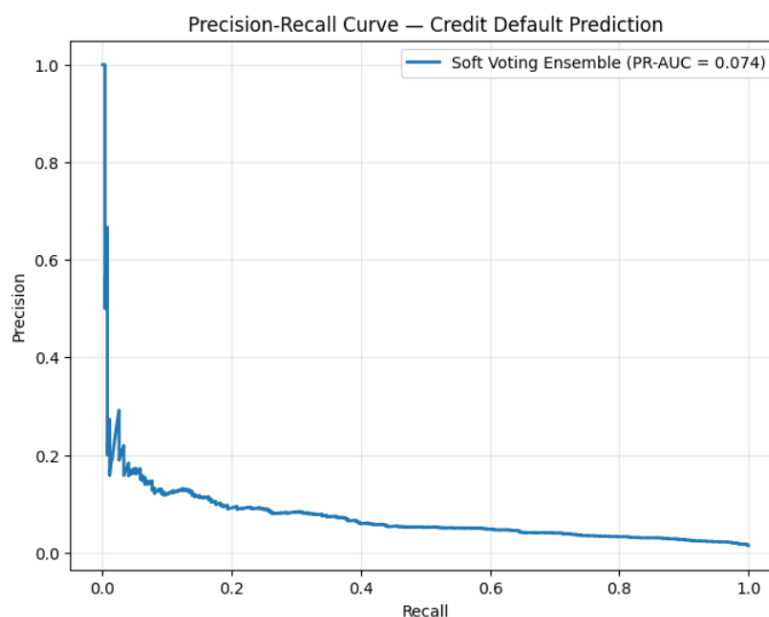


Figure 7: Precision-Recall curve of the final soft-voting ensemble.

11 Precision-Recall Performance of the Final Model

The final model achieves:

$$\text{PR-AUC} = 0.0736$$

The test-set default prevalence is approximately:

$$1.42\%$$

Thus, the PR-AUC is approximately $5.2\times$ the random-classification baseline implied by the default prevalence.

12 Threshold Optimization

The model produces a probability of default rather than an inherently binary decision. Therefore, the threshold at which an account is labelled “high risk” is a business decision.

The notebook evaluates thresholds from 0.01 to 0.50. At the final threshold of 0.50, the ensemble achieves:

- Precision = 3.84%
- Recall = 71.17%
- F1 = 0.0729

The threshold table demonstrates that reducing the threshold increases recall but reduces precision. This is exactly the type of trade-off that would need to be calibrated to the economics of a real credit portfolio.

For example:

- If the cost of missing a genuine defaulter is very high, the bank may prefer a lower threshold.
- If unnecessary interventions are expensive or damaging to customer experience, the bank may prefer a higher threshold.

Therefore, the model should not be deployed with a threshold selected purely because it is the conventional value of 0.50.

13 Final Confusion Matrix and Error Analysis

At the selected threshold of 0.50, the confusion matrix is:

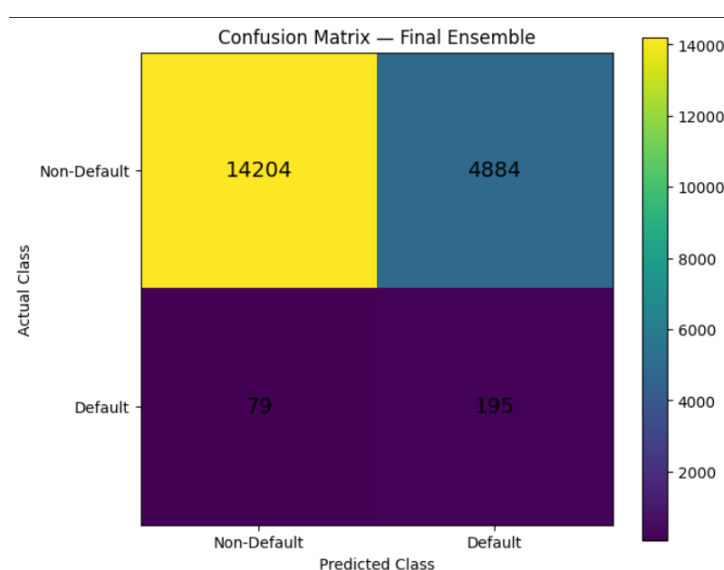


Figure 8: Confusion matrix of the final soft-voting ensemble at threshold = 0.50.

This gives:

- True Negatives = 14,204
- False Positives = 4,884
- False Negatives = 79
- True Positives = 195

The model identifies approximately:

$$195/274 = 71.17\%$$

of actual defaulters. However, it also flags:

$$4,884 \text{ non-default customers}$$

as potential defaulters. This illustrates the fundamental precision-recall trade-off.

14 Business Interpretation

The model should not be interpreted as a standalone automated credit decision engine. Instead, the strongest application of the current model is as a risk-ranking and early-warning layer.

For example, the model could assign each account a probability of default and divide the portfolio into risk bands:

Table 4: Proposed Risk Banding for Portfolio Action

Risk band	Potential use
Very low risk	Standard monitoring
Low risk	Standard servicing
Medium risk	Enhanced monitoring
High risk	Proactive customer outreach
Very high risk	Credit-policy review

This approach is preferable to treating the output as a binary approval/rejection signal because the model’s precision at the selected threshold remains low.

14.1 Cost-Sensitive Decision Making

The confusion matrix makes the business trade-off explicit. A false negative represents a customer who is actually going to default but is not flagged. A false positive represents a customer who is flagged despite not defaulting.

These errors do not have equal economic consequences. For a financial institution:

$$\text{Cost}(\text{False Negative}) \neq \text{Cost}(\text{False Positive})$$

Therefore, an ideal production implementation should incorporate explicit business costs into threshold optimization. A simple objective could be:

$$\text{Expected Cost} = FN \times \text{Cost}(FN) + FP \times \text{Cost}(FP)$$

The optimal threshold would then minimize expected cost rather than maximize accuracy. This would make the modelling framework substantially more aligned with real credit-risk decisioning.

15 Key Technical Learnings

15.1 Accuracy can be misleading

The non-SMOTE Logistic Regression model achieved 98.57% accuracy but 0% recall. This demonstrates that high accuracy is essentially meaningless when the positive class represents only 1.42% of the dataset.

15.2 Class imbalance must influence the modelling pipeline

SMOTE increased recall from 0% to more than 70% for the final ensemble. However, this improvement came with lower precision. This reinforces that imbalance treatment does not “solve” the problem; it changes the model’s operating trade-off.

15.3 Feature-family segmentation is useful

The dataset contains four structurally different feature families. Treating them separately before dimensionality reduction allows the modelling pipeline to preserve some of that structure.

15.4 Dimensionality reduction involves information trade-offs

Reducing 1,212 features to 20 components substantially simplifies the model. However, only 13.32% of transaction-feature variance is retained in the first five transaction components. Therefore, dimensionality reduction improves computational tractability at the cost of potentially discarding predictive information. This is one of the clearest areas for future improvement.

15.5 Model performance must be evaluated from multiple perspectives

The model’s ROC-AUC is approximately 0.81, but its precision at the 0.50 threshold is only 3.84%. A single metric cannot adequately characterize the usefulness of a credit-risk model. The evaluation should therefore include:

- ROC-AUC,
- PR-AUC,
- precision,
- recall,
- F1,
- confusion matrix,
- threshold analysis,
- top-risk segment performance,
- business cost.

16 Limitations

16.1 PCA interpretability

PCA transforms original variables into latent components. While this reduces dimensionality, it makes it difficult to explain exactly which customer behaviours drive an individual prediction. This is an important limitation for credit-risk applications where interpretability and model governance matter.

16.2 Fixed five-component allocation

Five components were retained from every feature family. However, the variance-retention results show substantial differences across families. Transaction features retain only 13.32% of variance, while bureau-enquiry features retain 69.31%. A family-specific component-selection strategy could therefore potentially improve predictive performance.

16.3 SMOTE may not perfectly reproduce credit-risk distributions

SMOTE generates synthetic minority observations. Although useful for improving minority-class learning, synthetic observations may not perfectly reflect the underlying distribution of genuine defaulters. Future iterations should compare SMOTE against:

- class-weighted models,
- focal loss,
- balanced boosting,
- cost-sensitive learning.

16.4 Threshold selection

The 0.50 threshold is not necessarily optimal for a bank. A production system should select a threshold using:

- expected loss,
- intervention capacity,
- portfolio risk appetite,
- customer impact,
- acceptable false-positive rate.

16.5 Probability calibration

A model's ranking performance and its probability calibration are different properties. A customer receiving a predicted probability of 0.40 should ideally correspond to an approximately 40% observed default rate among comparable customers. Future iterations should therefore evaluate:

- calibration curves,
- Brier score,
- Platt scaling,
- isotonic regression.

17 Future Modelling Roadmap

The next version of the model could improve the pipeline in five areas.

17.1 Replace blanket PCA with predictive feature selection

Instead of retaining five components from every feature family, compare:

- Mutual Information,
- model-based feature importance,
- recursive feature elimination,
- SHAP-based selection.

This would retain variables based on their relationship with default rather than their contribution to total variance.

17.2 Compare SMOTE against class-weighted boosting

A particularly important experiment would be:

SMOTE + classical models

versus

class-weighted XGBoost / LightGBM / CatBoost

This would determine whether synthetic oversampling is actually necessary.

17.3 Optimize for top-K risk detection

In a bank, the objective may be to identify the highest-risk 1%, 5% or 10% of customers. Therefore, future evaluation should include:

- Precision@1%
- Recall@1%
- Precision@5%
- Recall@5%
- Lift@1%
- Lift@5%

These metrics are closer to how a risk team might operationalize a model.

17.4 Cost-sensitive threshold optimization

Instead of using 0.50 by default, assign business costs to false positives and false negatives. Then determine:

$$\text{Threshold}^* = \text{argmin Expected Portfolio Cost}$$

This converts the ML output into a business decision framework.

17.5 Model explainability

For a production credit-risk model, the ability to explain predictions is critical. Future iterations should incorporate:

- SHAP values,
- feature-level risk drivers,
- partial dependence,
- individual customer explanation reports.

This would allow risk managers to understand not only who is high-risk, but why.

18 Conclusion

This project developed an end-to-end credit-card default prediction pipeline on 96,806 customer accounts and 1,212 modelling features. The primary challenge was the severe class imbalance, with only 1.42% of accounts classified as defaults.

The modelling pipeline addressed this through:

1. feature-family segmentation,
2. leakage-safe preprocessing,
3. standardization,
4. family-wise PCA,
5. training-only SMOTE,
6. multi-model benchmarking,
7. soft-voting ensemble modelling,
8. threshold analysis.

The final ensemble achieved:

$$\text{ROC-AUC} = 0.8088 \quad \text{and} \quad \text{PR-AUC} = 0.0736$$

while identifying approximately 71% of actual defaults at the selected 0.50 threshold.

More importantly, the project demonstrates that credit-risk modelling is not simply a model-accuracy problem. The analysis reveals a fundamental trade-off between identifying more genuine defaulters and limiting unnecessary interventions on good customers.

The strongest future direction is therefore not merely increasing model complexity, but integrating predictive modelling, probability calibration, threshold optimization, explainability and explicit business-cost modelling into a unified credit-risk decision framework.

The project ultimately demonstrates an end-to-end approach to converting a highly dimensional, highly imbalanced customer dataset into a practical risk-scoring framework while identifying the methodological and business considerations required before production deployment.